

Decomposing the effect of Ownership Changes on Nursing Home Staffing

Joseph White

Miami University

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Sources and Sample Selection

- Center for Medicare and Medicaid Services (CMS)
 - ▶ Nursing Home Compare (NHC) Archive
 - ▶ Healthcare Cost Report Information System (HCRIS)
 - ▶ Payroll Based Journal (PBJ)
- Facility-month-year level
- 48 Contiguous U.S. States over the period January 2017 to June 2024
- Freestanding facilities
 - ▶ Hospital based facilities make up 12% of all facilities in the U.S.

Identifying Ownership Changes - Background

- HCRIS ownership change flag
 - ▶ May reflect purely legal restructuring
- We validate using ownership data in the NHC archive
 - ▶ Reports every level of ownership, management through direct owners
 - ▶ Limited Liability Corporations
- We identify ownership changes at the highest level of ownership

Identifying Ownership Changes - Criteria

- ① Majority (50%) of ownership transfers from month $t - 1$ to month t
- ② Partial percentages we normalize and missing we assign equal weights across
- ③ We don't flag ownership changes when 80% stake remains in the same family or for changes in legal names (e.g., a change from "Smith Holdings LLC" to "Smith Family Partners LP")
- ④ Our sample then included facilities that facilities that:
 - ① Have no ownership changes in either dataset
 - ② Have exactly one change in each dataset within six months of each other
- ⑤ For these facilities we define the change month as the first month the NHC archive indicates a change

Dependent Variables - Nurse Staffing Levels

- The staff types we look at are RN's, LPN's, CNA's, and total staffing
- PBJ
- Hours per patient day (HPPD) for each staff type s , facility i , and period t

$$\text{HPPD}_{i,t} = \frac{H_{s,i,t}}{R_{s,i,t}}, \quad (1)$$

where $H_{s,i,t}$ is the total hours worked by the staff type s during month t for facility i and $R_{s,i,t}$ is the total resident days in that facility month

Dependent Variables - Nurse Staffing Levels

- We exclude a small number of observations to align with the implausible criteria outlined in the CMS NHC Technical Users Guide where:
 - 1 RN and LPN HPPD are both 0 or
 - 2 Total HPPD is below 1.5 or above 12 or
 - 3 CNA HPPD exceeds 5.25

Sample Construction

- Initial universe: all CMS-certified nursing homes
- Restrict to freestanding facilities
- Merge PBJ staffing, HCRIS, and NHC ownership data
- Drop implausible staffing observations (CMS PBJ rules)
- Final analytic sample:
 - ▶ 7,806 facilities
 - ▶ 632,231 facility-month observations

Summary Statistics